

Posters.science: Building Open Infrastructure for FAIR and AI-ready Scientific Posters

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Research Professor

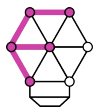
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Software Developer

Dorian Portillo, B.Sc.
Software Developer



CALIFORNIA MEDICAL
INNOVATIONS INSTITUTE



FAIR DATA
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Posters Are Important



Millions presented annually

One of the most common forms of scientific communication



Typically the first communication of a project

Early stage findings, presented first time outside the team



Can increase the pace of discoveries

If shared and made reusable



Poster findings are not always published

Studies find 1 to 50% of posters are published as papers
(takes ~2 years on average)

Posters Are Not Consistently Shared

Despite searching thoroughly, we found
only ~150,000 posters shared across 86 platforms

Metadata	Zenodo	Figshare
Title	100%*	100%*
At least one keyword	54%	100%*
Funding information	20%	28%
ORCID for all authors	26%	14%
At least one reference	8%	0%
Conference name	47%	NA
Conference dates	45%	NA

% of poster records containing structured metadata, * = mandatory
Zenodo: ~24,000 records labeled as posters, Figshare: ~14,000

There Are Many Challenges to Poster Sharing



Lack of awareness (why/how to share posters?)



Time consuming & no incentives



Unstructured (programmatic reuse is difficult)

Posters.science: Streamlined Poster Sharing

Posters.science is an **open-source and free platform** for easily sharing FAIR and AI-ready posters on Zenodo in 3 steps



1

Upload Poster

Metadata (authors, affiliation, etc.) auto-extracted with poster2JSON tool



2

Review & Edit

Convenient user interface provided with integrations (ORCID, ROR, etc.)



3

Share on Zenodo

Poster file and machine actionable poster.json archived with rich metadata

Posters.science: Machine Actionable Posters



Toward the Envision Portal: Designing a FAIR and AI-Ready Framework for Ophthalmic Imaging Sharing and Discovery

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Xuebin Dong, MS¹, Nahid Zenati, PhD², Cecilia Lee, MD, MS²

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4000



Background

Ophthalmic imaging datasets, such as optical coherence tomography (OCT) and fundus photographs, are a valuable resource for advancing AI/ML models for disease detection. However, they are not consistently shared. When shared, their reuse is hindered by fragmented sharing that limits findability and by inconsistent structure that reduces interoperability and machine usability (AI/ML).

Purpose

We are developing the **Envision Portal (EP)**, an open source platform for easily sharing, finding, and reusing ophthalmic imaging data. We are currently focusing on two major features: **Automated** to format ophthalmic imaging data into FAIR (Findable, Accessible, Interoperable, Reusable) and machine-usable datasets

Methods

- We reviewed FAIR data principles and ophthalmic imaging standards, including guidelines from NIH Bridge2AI.
- We developed a SelfFit few-shot classifier trained to classify dataset metadata (title, description, keywords, and file type) as ophthalmic imaging dataset or not. The classifier is built on the all-mpnet-base-v2 sentence transformer (768-dim) and was trained on 891 curated examples (262 positive, 629 negative).

Results

We finalized major design decisions and established the base framework of the Envision Portal:

- The Clinical Dataset Structure (CDS) will be used for creating standard format datasets (Figure 1).
- DICOM will be used as the preferred format for imaging data.
- The SelfFit classifier achieved a test accuracy of 0.981 and an F1 of 0.935 on the positive class. Expert validation is currently ongoing.

Applied on datasets from six repositories, the pipeline (Figure 2) identified 1800+ ophthalmic imaging datasets that can be browsed on the platform at envisionportal.org

Conclusion

Reviewing existing imaging standards allowed us to identify the key components a standardized ophthalmic imaging dataset should contain, leading to the adoption of CDS and DICOM as the foundation for the Envision Portal.

Our SelfFit-based classifier demonstrates that cost-effective ophthalmic imaging dataset discovery is achievable from metadata alone, without accessing underlying image files. This will be further validated through an expert curated training set.

Future work will focus on expanding repository coverage for dataset discovery and developing tools to format any set of ophthalmic data files into our prescribed dataset format.

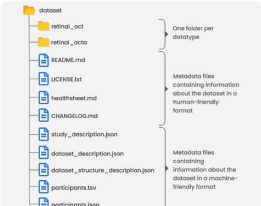


Figure 1. Illustration of a dataset organized according to the Clinical Dataset Structure (CDS) developed by the AI-READY project in the NIH Bridge2AI program.

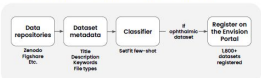


Figure 2. Illustration of the structure of the pipeline to identify ophthalmic datasets based on their metadata.

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envisionportal.org
fairdatahub.org

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Find this poster and all related references here



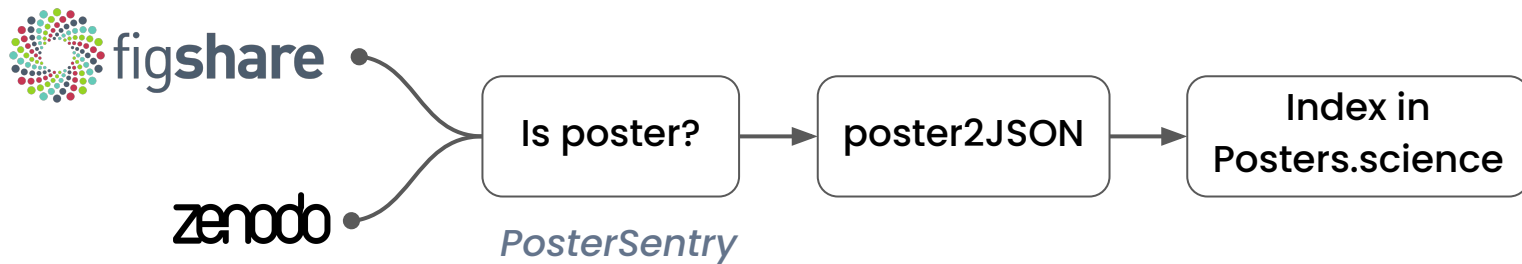
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Auto-generated poster.json file following our JSON schema for poster inspired by the DataCite metadata schema

Posters.science: Poster Indexing and Discovery

Posters.science also **indexes posters from different platforms** to make them conveniently findable in one place



Current Status

Progress



Posters.science is live!



24,000+ poster indexed

Next Steps



Index more posters



Promote Posters.science

Stop by our poster!

Join us at CoFest!

Thank You!

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